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MILITARY SPECIFICATION

INDICATOR, VOLTAGE: BATTERY-GENERATOR
28 VOLT DC (NOMINAL)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and inspections of a battery-generator indicator, referred to herein as "indicator," utilizing a 28 volt (V) (nominal) direct current (dc) electrical system.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 6625

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SPECIFICATIONS
FEDERAL

L-P-380	- Plastic Molding Material Methacrylate.
L-P-393	- Plastic Molding Material, Polycarbonate, Injection and Extrusion.

MILITARY

MIL-W-80	- Window, Observation, Acrylic Base, Antielectrostatic, Transparent (for Indicating Instrument).
MIL-E-52798	- Enamel, Alkyd, Camouflage.
MIL-E-52835	- Enamel, Modified Alkyd, Camouflage, Lusterless.
MIL-P-81390	- Plastic, Molding Material, Polycarbonate, Glass Fiber Reinforced.

STANDARDS
FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-193	- Painting Procedures and Marking for Vehicles, Construction Equipment and Material Handling Equipment.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-1184	- Electrical Components for Automotive Vehicles; Waterproofness Tests.
MIL-STD-45662	- Calibration Systems Requirements.
MS24532	- Voltmeter: Battery-Generator 28 Volt DC.

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D256 - Impact Resistance of Plastics and Electrical Insulating Materials.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), a first article of indicators to be furnished under this specification shall be produced prior to the manufacture of the items in production quantities. Voltage indicators furnished under this specification shall be products which are represented by a first article sample that has been tested and has passed inspection specified herein (see 4.4 and 6.3).

3.2 Materials. Materials shall be as specified herein, on applicable standards or drawings, and in applicable specifications (see 4.8.1).

3.2.1 Metals. Exterior metals shall be of a corrosion-resistant type or shall be suitably treated to resist corrosion (see 4.8.1).

3.2.2 Case. The case shall be constructed of metal suitable for the purpose or plastic molding material conforming to MIL-P-81390, type II or III (see 4.8.1).

3.2.3 Cover window material. The cover window material shall be treated on both sides with an electrostatic solution unless nonsusceptibility to static generating characteristics has been demonstrated by the procedure specified herein (see 3.4.5 and 4.8.11). The window thickness shall be 0.125 $\pm$ 0.015 inches [3.18 $\pm$ 0.38 mm (millimetre)]. The window shall demonstrate an Izod strength of not less than 0.3 ft. lb (foot pounds) per inch of notch [16 J (joule) per metre of notch] when tested as specified by procedure C of ASTM D256. The window material shall conform to MIL-W-80 or, if molded, to that of type I, class 3 of L-P-380 or L-P-393 (see 4.8.1 and 4.8.2).

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3.2.4 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 6.4.1).

3.3 Design and construction. Indicators shall be constructed to the form and dimensions specified on MS24532 (see 4.8.1 and 4.8.2).

3.3.1 Standard parts. Military standard parts shall be used wherever applicable. Commercial standard parts may be used provided they are interchangeable without modification with the military standard parts they replace (see 4.8.1 and 4.8.2).

3.3.2 Threaded parts. All screw threads shall conform to FED-STD-H28 (see 4.8.1 and 4.8.2).

3.3.3 Locking devices. Lockwashers, self-locking nuts, or other approved locking devices shall be incorporated where specified or where required to prevent loosening of components (see 4.8.1 and 4.8.2).

3.3.4 Dial.

3.3.4.1 Dial marking. The face of the dial shall be marked as shown on MS24532 (see 4.8.1).

3.3.4.2 Fastening. The dial shall be secured to the body of the indicator in such a position that the marking will be legible when illuminated through the case windows from an outside source. The method of securing the dial shall be such that it will not become loose or turn when subjected to vibration (4.8.7.1) or shock (4.8.7.2).

3.3.4.3 Dial illumination. Indicators shall provide side windows or other provisions for adequate indirect illumination of dial in accordance with MS24532 (see 4.8.1).

3.3.5 Pointer. The pointer shall be damped to prevent oscillation of no more than 2 degrees at all indicated points within the range of the indicator, when subjected to vibration and shock (see 4.8.7.1).

3.3.6 Cover window. The window shall be secured to the base excluding moisture from the instrument case (see 4.8.2).

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3.4 Performance.

3.4.1 Calibration. When tested as specified, calibration shall be as follows (see 4.8.3):

a. With the pointer positioned midway between the left and right limits of the indentation in the top of the green section of the generator arc, the applied voltage shall be not less than 28 V nor more than 29 V, except that after testing as specified in 4.8.5, 4.8.7.1, 4.8.7.2 and 4.8.4, the applied voltage shall be not less than 27.5 nor more than 29.5 V.

b. With the pointer positioned at the midpoint between the low voltage red and the yellow sections of the battery arc, the applied voltage shall be not less than 21 V nor more than 23 V, except that after testing as specified in 4.8.5, 4.8.7.1, 4.8.7.2 and 4.8.4, the applied voltage shall be not less than 20 V nor more than 24 V.

3.4.2 Temperature influence. The indicator shall operate throughout an ambient air temperature range of -65 °F (-54°C) to + 160°F (+ 71°C). When tested as specified, the indicator shall conform to 3.4.1 (see 4.8.4).

3.4.3 Endurance. When mounted and calibrated as specified in 4.8.3, the indicator shall be subjected to full scale deflection and return for a total of 200,000 cycles. After testing, the indicator shall conform to 3.4.1 (see 4.8.5).

3.4.4 Dielectric withstanding voltage. The insulation of the metal case construction indicator shall show no evidence of cracking or charring when tested as specified in method 301 of MIL-STD-202 (see 4.8.6).

3.4.5 Electrostatic susceptibility. Indicators will demonstrate nonsusceptibility to static change generation when subject to the procedure specified in 4.8.11 (see 4.4 and 4.6).

3.5 Environmental.

3.5.1 Vibration. The indicator shall meet the requirements of 3.3.5 when subjected to the vibration test specified in method 201 of MIL-STD-202 and shall evidence no disintegration of component materials or assemblies. The specimen shall subsequently meet the requirements of 3.4.1 (see 4.8.7.1).

3.5.2 Shock. The indicator shall be subjected to the shock test as specified in method 213, test condition I of MIL-STD-202 obtaining an impact of 100G. The specimen shall evidence no disintegration of component materials or assemblies. The specimen shall subsequently meet the requirements of 3.4.1 (see 4.8.7.2).

3.5.3 Waterproofness. The indicator shall meet the waterproofness requirements as specified for type II, class 2 components of MIL-STD-1184 (see 4.8.8).

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3.5.4 Salt fog (corrosion). When tested as specified in method 509 of MIL-STD-810, the indicator shall evidence no corrosion affecting operation and shall subsequently meet the requirements of 3.4.1 (see 4.8.9).

3.5.5 Fungus. When tested as specified in method 508 of MIL-STD-810, the indicator shall evidence no fungus growth affecting operation and shall subsequently meet the requirements of 3.4.1 (see 4.8.10).

3.6 Finish (see 4.8.2).

3.6.1 Bezel. The indicator bezel shall be lusterless camouflage forest green color conforming to MIL-E-52798 or MIL-E-52835.

3.6.2 Pointer. The visible portion of the pointer shall be all white and non-luminous.

3.6.3 Preparation and painting. Pretreatment, priming and painting shall conform to the requirements of MIL-STD-193 except that, in no case, shall paint or other coating be applied to that portion of the case body so as to interfere with the intended purpose of dial illumination (see 3.3.4.3).

3.7 Identification marking. Identification marking shall be in accordance with MIL-STD-130 and as a minimum shall include the following (see 4.8.2).

Military part number
Manufacturer's name or identification
Date of manufacture (month, year e.g. June 87)
US

3.8 Workmanship. The workmanship shall be such as to assure a product free of rust, burrs, chips, sharp edges and scratches. All electrical connections shall be mechanically secured before being soldered or brazed, to preclude the possibility of becoming loose due to vibration or shock. All solder cleaning agents and flux shall be non-corrosive. All traces of flux shall be chemically neutralized and removed after soldering (see 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling for quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10% of the measurement tolerance. Accuracy of electrical measuring instrument shall be $\pm$ 0.50 percent (maximum) of measurement reading. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Tests (see 4.5.3).
c. Control tests (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all test specimens shall be temperature stabilized for one hour and inspections shall be conducted under the following conditions:

a. Air temperature 73 $\pm$ 18°F (23 $\pm$ 10°C)
b. Barometric pressure $28.5 \begin{smallmatrix} +2 \\ -3 \end{smallmatrix}$ inches of mercury ($725 \begin{smallmatrix} +50 \\ -75 \end{smallmatrix}$ mm Hg)
c. Relative humidity 50 $\pm$ 30%

4.3.1 Apparatus. Apparatus shall include a testing chamber capable of maintaining specified air temperatures within a range of - 70°F (-56.8°C) to + 165°F (73.8°C), a laboratory type dc voltmeter with accuracy measuring within $\pm$ 0.25%, a precision protractor, and a source of dc electrical energy. There shall also be a submersion tank capable of producing and maintaining both a pressure and a vacuum of 6 psi (pounds per square inch) [0.41 kPa (kilopascals)].

4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select a sample of three indicators from the first 10 units produced that shall be inspected in the order specified in table I and shall be subjected to the examinations and tests specified in table II. Approval

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of the first article sample by the Government shall not relieve the contractor of his obligation to supply indicators that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

TABLE I. Order of first article testing.

Specimen no.	Paragraph no.	Test sequence
1	4.8.3	Calibration
	4.8.11	Electrostatic susceptibility
	4.8.4	Temperature influence
	4.8.5	Endurance
2	4.8.3	Calibration
	4.8.7.2	Shock
	4.8.8	Waterproofness
	4.8.10	Fungus
3	4.8.6	Dielectric withstanding voltage
	4.8.3	Calibration
	4.8.7.1	Vibration
	4.8.9	Salt fog (corrosion)

TABLE II. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance		
				Examination	Tests	Control
Materials and construction	3.2 thru 3.2.3, 3.3 thru 3.3.4.1 and 3.3.4.3	4.8.1	X			
Defects (see 4.5.2 and table III)	3.3 thru 3.3.3, 3.3.4.2, 3.3.5, 3.3.6, 3.6, 3.7, and 3.8	4.8.2	X	X		
Calibration	3.4.1	4.8.3	X		X	X
Temperature influence	3.4.2	4.8.4	X			
Endurance	3.4.3	4.8.5	X			
Dielectric withstanding voltage	3.4.4	4.8.6	X			X

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TABLE II. Classification of inspections - Continued.

Title	Requirement	Inspection	First article	Quality conformance		
				Examination	Tests	Control
Electrostatic susceptibility	3.4.5	4.8.11	X			X 1/
Vibration	3.5.1	4.8.7.1	X			
Shock	3.5.2	4.8.7.2	X			
Waterproofness	3.5.3	4.8.8	X		X	X
Salt fog (corrosion)	3.5.4	4.8.9	X			
Fungus	3.5.5	4.8.10	X			

1/ Test is applicable only when an electrostatic coating is required in meter construction.

4.4.1 First article inspection failure. Test item deficiencies during, or as a result of, the first article test shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of, the first article test shall be evidence that all items already produced prior to completion of the first article test are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government shall not accept products until first article testing is completed to the satisfaction of the Government.

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the indicators of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.1.3 Sampling for acceptance tests. Samples for acceptance tests shall be selected in accordance with level S-3 of MIL-STD-105.

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4.5.2 Examination.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL).

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table III.

TABLE III. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 1.0% Defective	
101	Incorrect dimensions affecting interchangeability (see 3.3.1 and 3.3.2).	Gage
102	Nonconformance in design and construction (see 3.3).	Visual, tactile or SIE 1/
Minor	AQL 2.5% Defective	
201	Incorrect dimensions not affecting interchangeability (see 3.3 and 3.3.6).	Gage
202	Improper finish (see 3.6).	Visual
203	Improper marking (see 3.7).	Visual
204	Faulty workmanship (see 3.3.3, 3.3.4.2 and 3.8).	Visual

1/ SIE Standard Inspection Equipment.

4.5.3 Tests. Samples selected in accordance with 4.5.1.3 shall be subjected to the quality conformance test order specified in table IV, and shall conform to an AQL of 6.5.

TABLE IV. Order of quality conformance testing.

Paragraph No.	Test Sequence
4.8.3	Calibration
4.8.8	Waterproofness

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4.6 Control tests. Control tests shall be conducted on two from each lot of 500 units consecutively produced, except that not more than 4 tests shall be performed in a 2 month period, nor less than one test in a one month period. The indicators shall be subjected to the tests specified in table II in the order listed in table V.

TABLE V. Order of control testing.

Sample No.	Paragraph No.	Test Sequence
1 and 2	4.8.11	Electrostatic susceptibility
1 and 2	4.8.6	Dielectric withstanding voltage
1 and 2	4.8.3	Calibration
1 and 2	4.8.8	Waterproofness

4.7 Failure. Failure of any indicator to pass any of the specified inspections shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.8 Methods of inspection.

4.8.1 Materials and construction. Conformance to 3.2 through 3.2.3, 3.3 through 3.3.4.1, and 3.3.4.3 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.8.2 Defects. Conformance to 3.3 through 3.3.3, 3.3.4.2, 3.3.5, 3.3.6, 3.6, 3.7 and 3.8 shall be determined by examination for the defects listed in table III. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.8.3 Calibration. To determine conformance to 3.4.1, the indicator shall be mounted in the chamber in the vertical position. The applied voltage shall be adjusted so that the pointer is centered on one calibration point (see 3.4.1) and voltage recorded. The procedure shall be repeated, with voltage being recorded at the other calibration point.

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4.8.3.1 Attitude error. To determine conformance to 3.4.1, the indicator shall be mounted in such a manner that the dial face can be positioned as follows:

a. 45 degrees clockwise from the vertical plane.
b. 45 degrees counterclockwise from the vertical plane.
c. Face up (horizontal plane).
d. Face down (horizontal plane).

4.8.4 Temperature influence. To determine conformance to 3.4.2, the indicator shall be subjected to the test specified in 4.8.3, except that the ambient air temperature shall be stabilized at -65 ± 5°F (54.0 ± 2.8°C) and subsequently at + 160 ± 5°F (71.0 ± 2.8°C). Readings shall be taken at each temperature after four hours exposure.

4.8.5 Endurance. To determine conformance to 3.4.3, full scale deflection shall be caused by application of maximum voltage utilizing an "on-off" switch. One cycle of operation shall consist of one complete oscillation from extreme left to extreme right and return to extreme left. Cycling shall be at a rate as to produce full scale deflection of the indicator pointer.

4.8.6 Dielectric withstanding voltage. To determine conformance to 3.4.4, the indicator shall be tested as specified in method 301 of MIL-STD-202 except voltage shall be 440 V root mean square (rms) at a frequency of 60 cycles per second (cps), applied between the case and ungrounded terminal or winding for no less than 1 minute. This test is to be conducted prior to final assembly for sealed items.

4.8.7 Environmental tests.

4.8.7.1 Vibration. To determine conformance to 3.5.1, the indicator shall be subjected to the vibration test specified in method 201 of MIL-STD-202. After the test, the unit shall be checked for looseness and turning of dial and then subjected to the test specified in 4.8.3.

4.8.7.2 Shock. To determine conformance to 3.5.2, the indicator shall be mounted to simulate actual mounting in use. The test shall be repeated 10 times and applied in the vertical axis while the indicator is in its normal mounting orientation. An optional method, as specified in 4.8.7.3, may be used in such manner that an impact of 50 G will be obtained.

4.8.7.3 Shock, optional test procedure.

4.8.7.3.1 Apparatus. The shock-testing apparatus specified herein and shown in figures 1 to 6 inclusive, shall be used. The main structure consists of a cast-iron base to which upright channels are securely bolted. The tops of the upright channels are secured to a top channel which also supports rods to guide the movable carriage assembly. The other ends of the guide rods are screwed into the base. The carriage assembly carries on its underside a stiff calibrated spring resiliently mounted, but at the same time

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having helical compression springs sufficiently strong to hold the calibrated spring in contact with the effective cylindrical bearings. The cast-iron base shall be single-piece constructed so that it has no practical yielding. The prime requisite in the construction of the movable carriage assembly is rigidity. For this reason 0.250 inch-thick steel shall be used. After being securely fastened together with screws, the carriage parts shall be welded to give the equivalency of single-piece construction.

4.8.7.3.2 Mounting accessories. Mounting accessories which will assure rigid fastening of the specimens to the carriage assembly shall be used. Since special tests may require special mounting fixtures, the carriage test plate shall be provided with a number of extra holes. The top of the carriage frame shall be provided with a holding bolt to facilitate the rigid clamping necessary.

4.8.7.3.3 Stationary anvil. The stationary anvil shall be made of hardened tool steel with a radius curvature of 1.50 inches and shall be bolted tightly to the base.

4.8.7.3.4 Calibrated spring. The calibrated spring shall be made of first-quality carbon tool steel, spring-tempered, and having a hardness of 85 to 95, Rockwell B scale. It shall be annealed and ground flat stock of the following composition:

Carbon	1.08 to 1.18
Chromium	0.40 to 0.50
Manganese	.20 to .30
Silicon	.20 to .30
Phosphorus and sulfur	.025 maximum

The spring shall be 1.75 inches wide and approximately 0.150 inch thick, and shall have an overall length of 7.5 inches. The approximate spring constant (K) shall be between 5000 and 5500 pounds-per-inch deflection.

4.8.7.3.5 Calibration. The actual calibration shall be made on the completely assembled machine. Static loads up to 500 pounds shall be applied to the top of the carriage frame, and the spring deflections measured by means of a dial gage or height gage. The gage shall be mounted on the top surface of the cast-iron base as one reference surface, and measurements shall be taken at the center of the carriage test plate.

$$\text{Spring constant (K)} = \frac{\text{load in pounds}}{\text{inches deflection}}$$

For a spring constant of 5000, the deflection of a 500-pound load would be 0.100 inch.

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4.8.7.3.6 Calibration curves. Use of the shock-testing mechanism is facilitated by plotting calibration curves for various weights of specimens. Ordinarily, parallel curves for specimen weights of 0.5 pounds and 2.0 pounds will be adequate, and intermediate weights can be estimated. Figure 6 shows such curves arbitrarily calculated on the basis of a mean spring constant (K) of 5250 and a total carriage-and-mounting-fixture weight of 9.75 pounds.

4.8.7.3.7 Computation of acceleration (G). The following formula for obtaining the required acceleration or shock, in terms of gravity units (G), is utilized for variations of component weight and spring constant (K):

$$G = \sqrt{\frac{2hK}{w}}$$

When G is specified, the necessary height may be determined from:

$$h = \frac{G^2 w}{2K}$$

Where:

G = nominal value of acceleration (gravity units).
h = height from which carriage is dropped (inches).
K = spring constant (pounds-per-inch deflection).
w = weight of carriage and its test load (pounds).

4.8.7.3.8 Procedure. The shock-testing apparatus shall be mounted on a sturdy laboratory table or equivalent base and leveled before use. The specimen shall be rigidly mounted on the carriage. For each blow the carriage shall be raised to the height necessary for obtaining the required acceleration and then allowed to fall. Means shall be provided to prevent the carriage from striking the anvil a second time.

4.8.8 Waterproofness. To determine conformance to 3.5.3, the indicator shall be subjected to the waterproofness test as specified in procedure I of MIL-STD-1184 to determine component resistance to the effects of water.

4.8.9 Salt fog (corrosion). To determine conformance to 3.5.4, the indicator shall be subjected to the salt fog test specified in method 509 of MIL-STD-810, except that the duration of the test shall be 200 hours. After the test, the unit shall be subjected to the test specified in 4.8.3.

4.8.10 Fungus. To determine conformance to 3.5.5, the indicator shall be subjected to the fungus test specified in method 508 of MIL-STD-810, except that the test duration shall be continuous for 90 days. After the test, the unit shall be subjected to the test specified in 4.8.3.

4.8.11 Electrostatic susceptibility. Specimens are to be conditioned at 73 $\pm$ 5°F (23 $\pm$ 2.8°C) and 15 $\pm$ 5 percent relative humidity for a minimum of 8 hours immediately prior to test specified herein. Pointer will not be deflected by a static charge generated when the window is rubbed with a cloth containing not less than 95% fleece or pulled wool (see 3.4.5).

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5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The indicator procured by this specification measures and visually displays the system voltage on automotive, marine and industrial engines utilizing a 28 V (nominal), dc electrical system.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Applicable military drawing, revision level and date (see 3.3).
c. First article requirements if not as specified (see 3.1).
d. Responsibility for inspection if other than as specified herein (see 4.1).
e. Responsibility for inspection equipment if other than as specified herein (see 4.1.2).
f. Inspection conditions if other than as specified herein (see 4.3).
g. Selection of applicable level and packaging requirements (see 5.1).

6.3 First article. First article samples shall be tested and approved under the appropriate provisions of sections 52.209-3 and 52.209-4 of the Federal Acquisition Regulations. The contracting officer should include specific instructions in all approval instruments regarding arrangements for examination, test, and acquisition of the first article (see 3.1).

6.4 Definitions.

6.4.1 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.4.2).

6.4.2 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

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6.5 Subject term (key word) listing.

Battery - generator indicator
Fungus resistant
Indicator
Low temperature compatible
Meter, arbitrary scale
Meter, special scale
Shock, resistant
Vibration resistant
Voltage indicator
Voltmeter, battery-generator

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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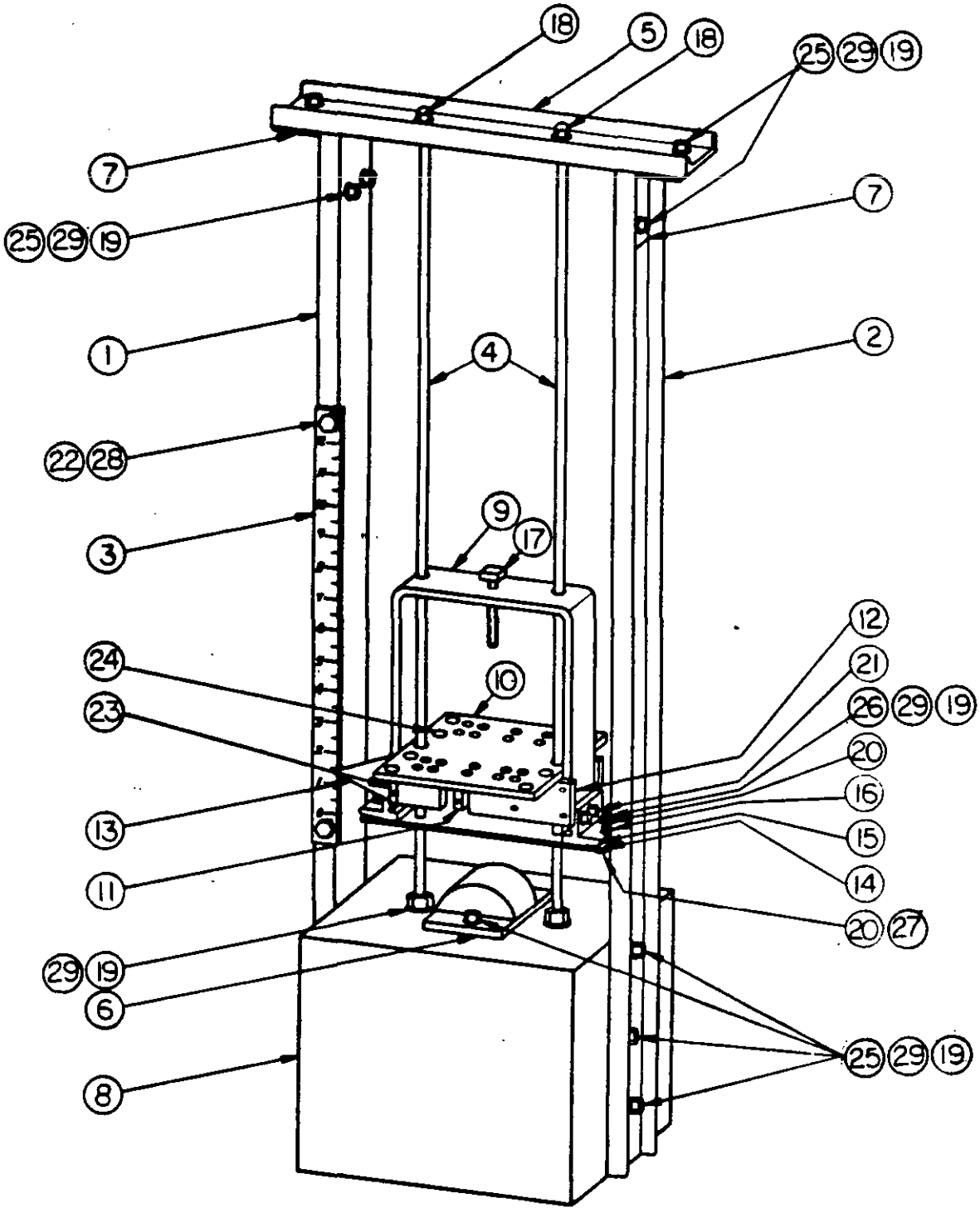


FIGURE 1. General assembly of shock-testing apparatus (see fig. 5).

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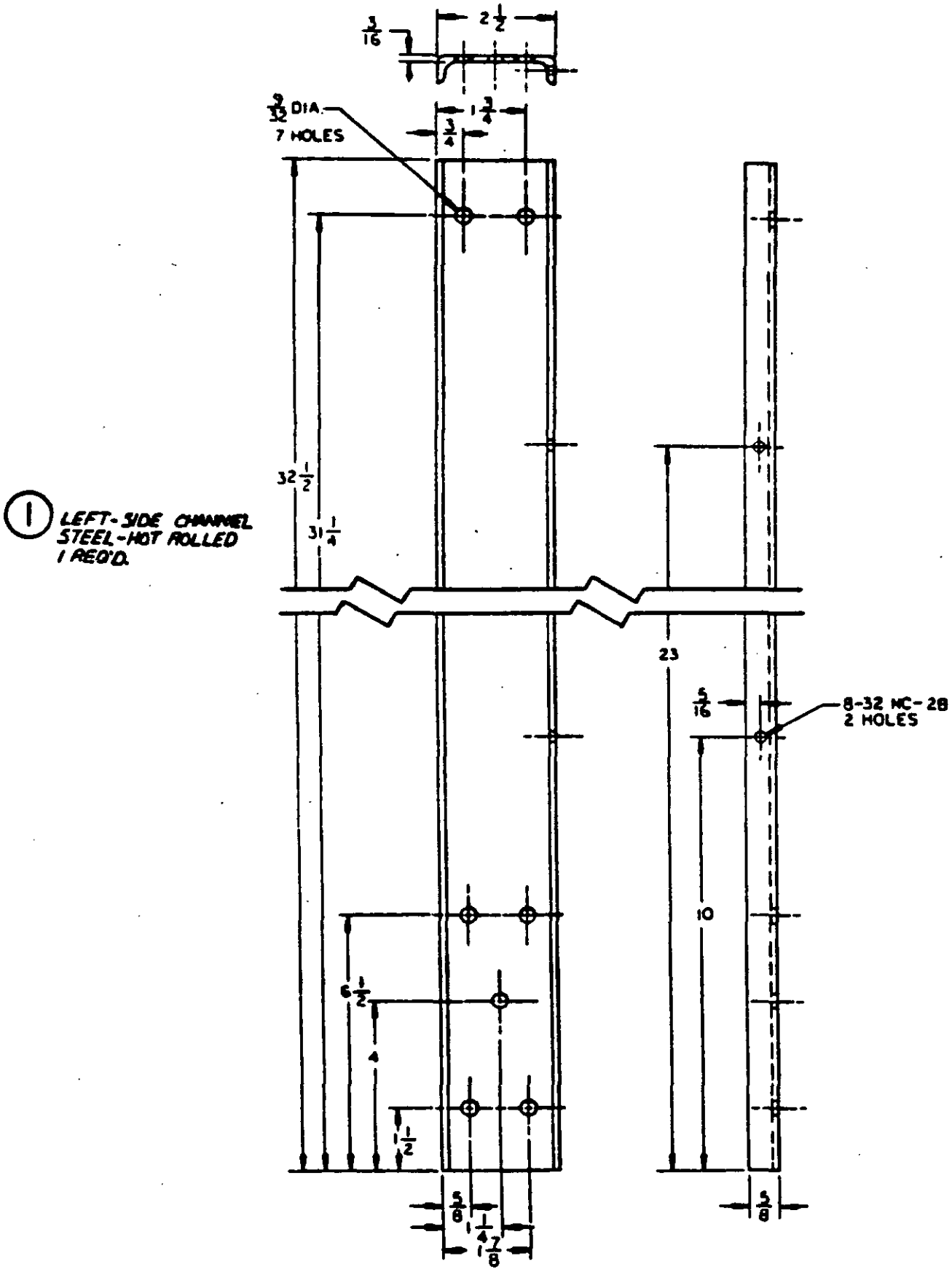


FIGURE 2. Main frame details (see fig. 5).

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(2) RIGHT-SIDE CHANNEL STEEL - HOT ROLLED 1 REQ'D.

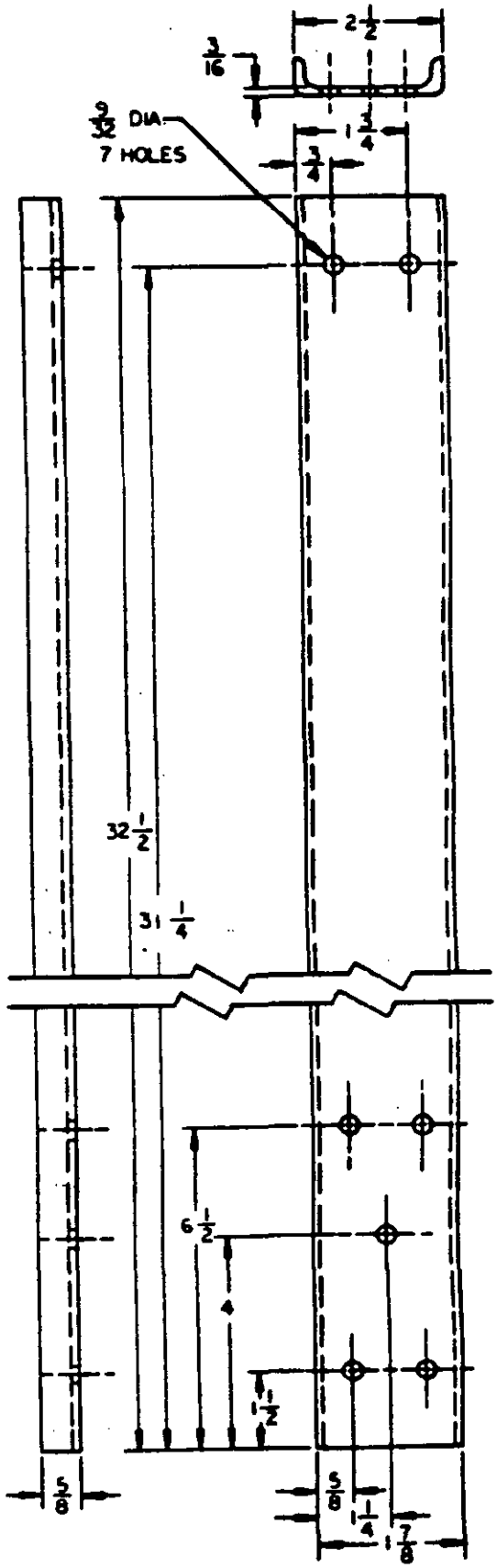


FIGURE 2. Main frame details (see fig. 5). -cont'd

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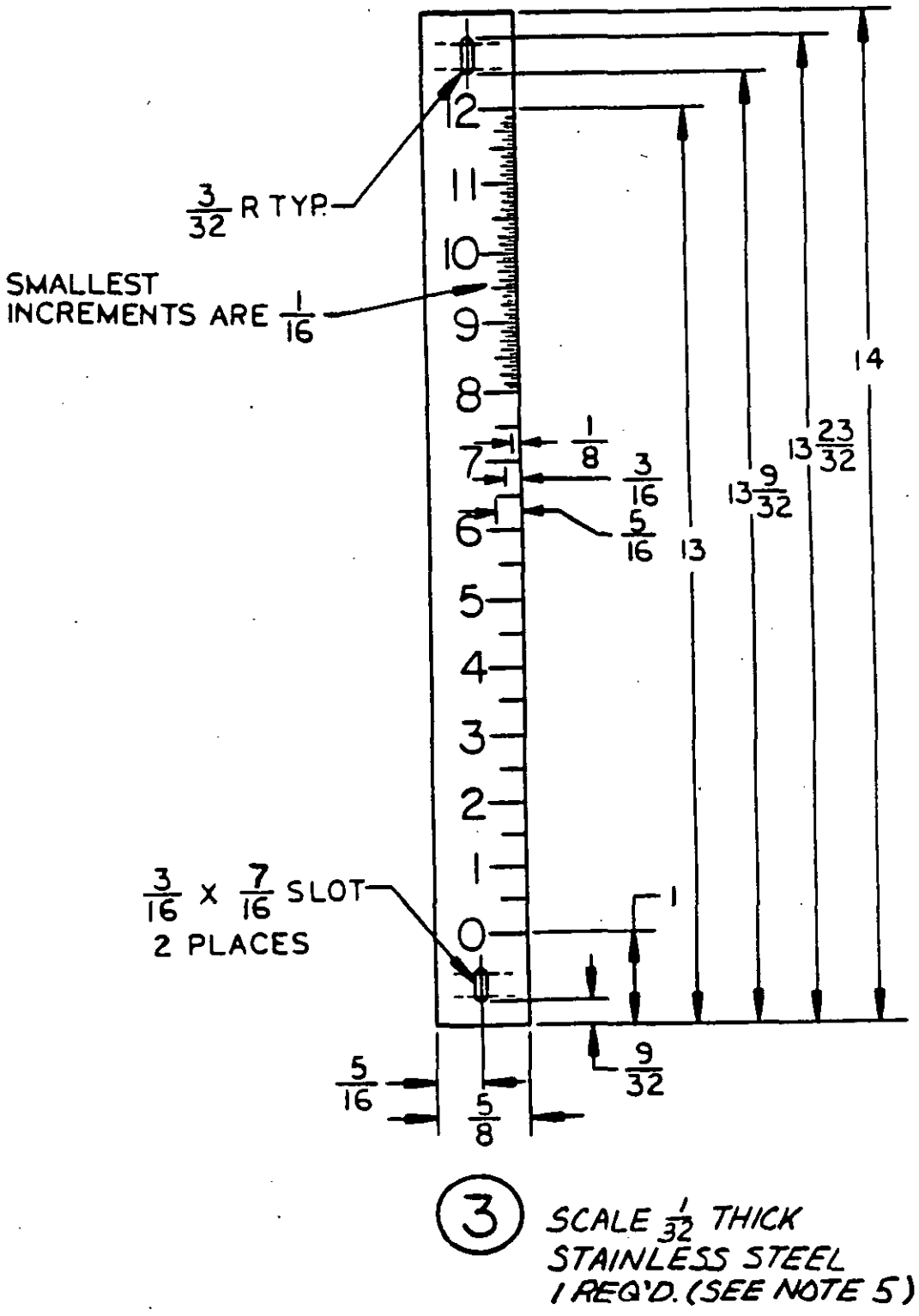
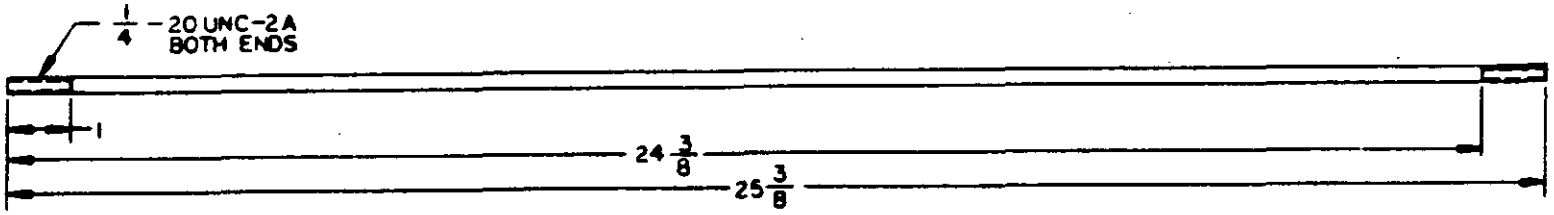
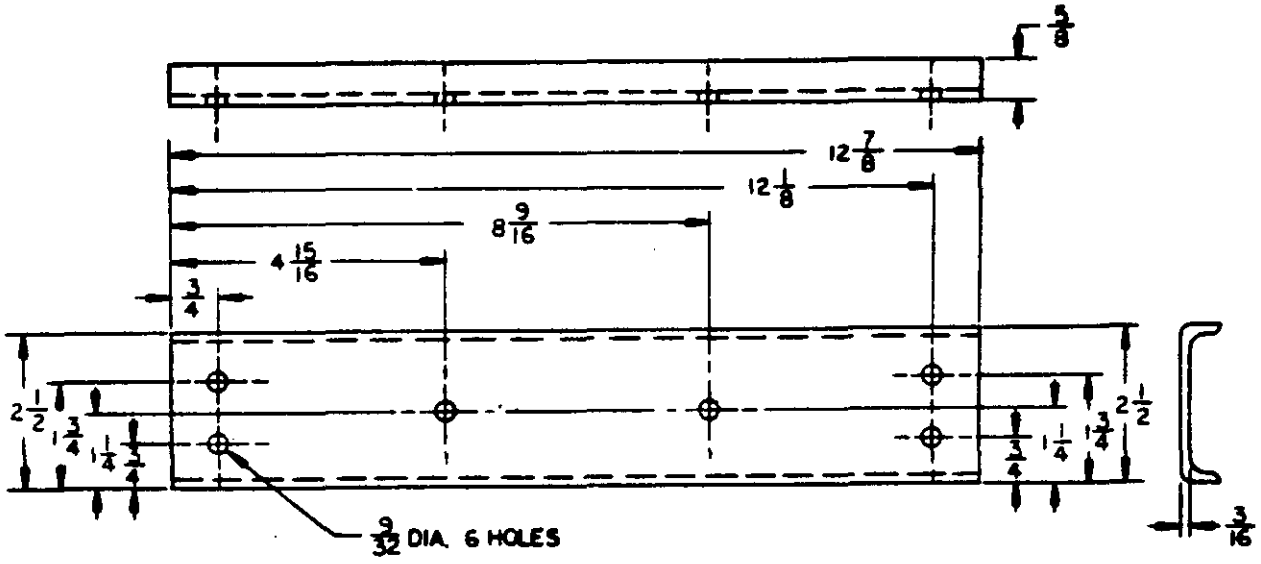


FIGURE 2. Main frame details (see fig. 5). -cont'd

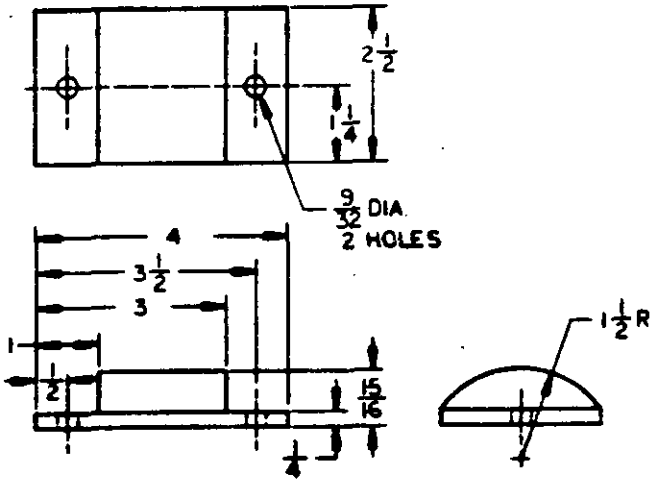
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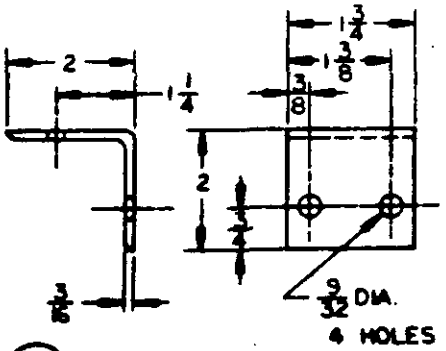
4 GUIDE ROD
STEEL
2 REQ'D.



5 TOP CHANNEL
STEEL - HOT ROLLED
1 REQ'D.



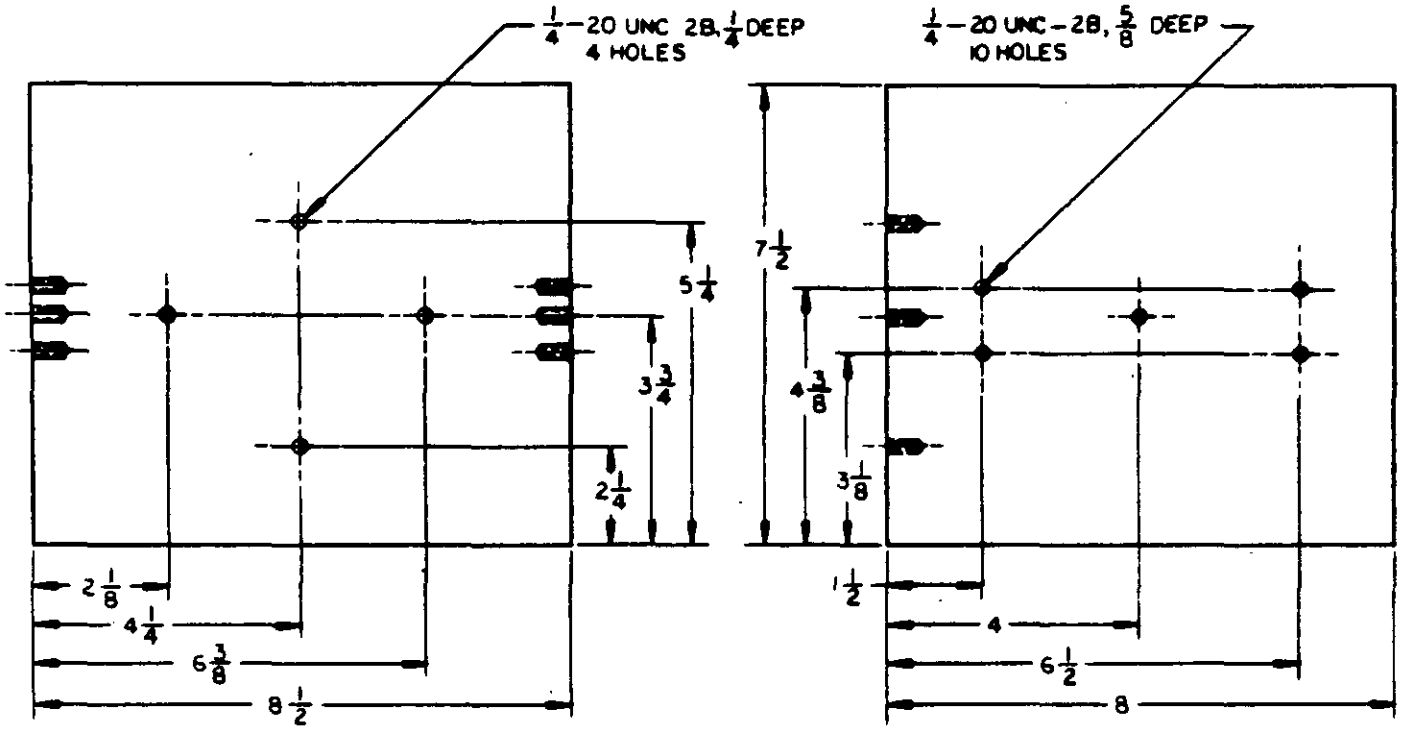
6 ANVIL
TOOL STEEL - HARDEN & DRAW 60-62
1 REQ'D. ROCKWELL C



7 TOP ANGLE
STEEL - HOT ROLLED
2 REQ'D.

FIGURE 2. Main frame details (see fig. 5). -cont'd

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(8) BASE
CAST IRON - NORMALIZED
1 REQ'D.

FIGURE 2. Main frame details (see fig. 5) -cont'd

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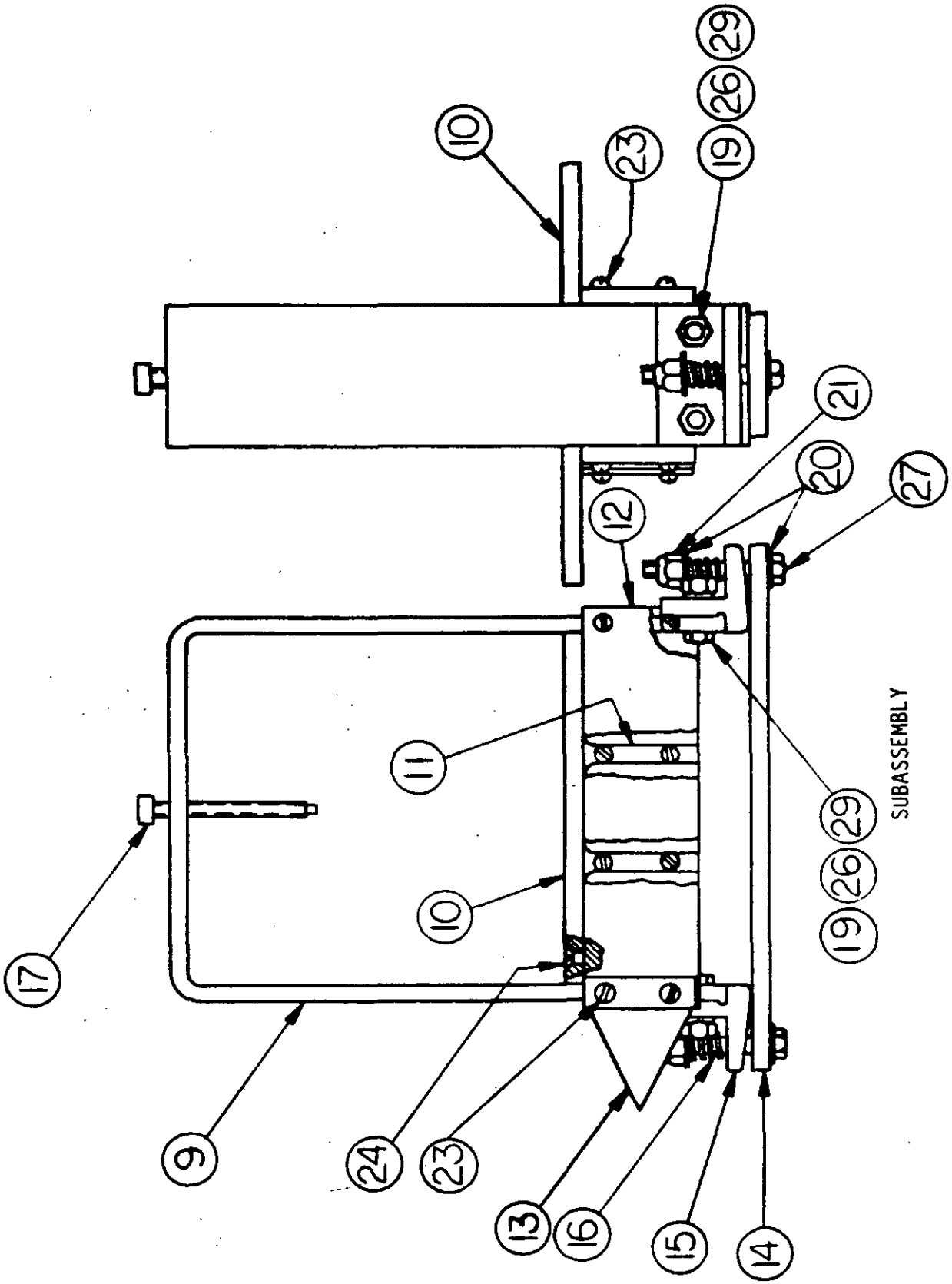


FIGURE 3. Carriage-assembly (see fig. 5).

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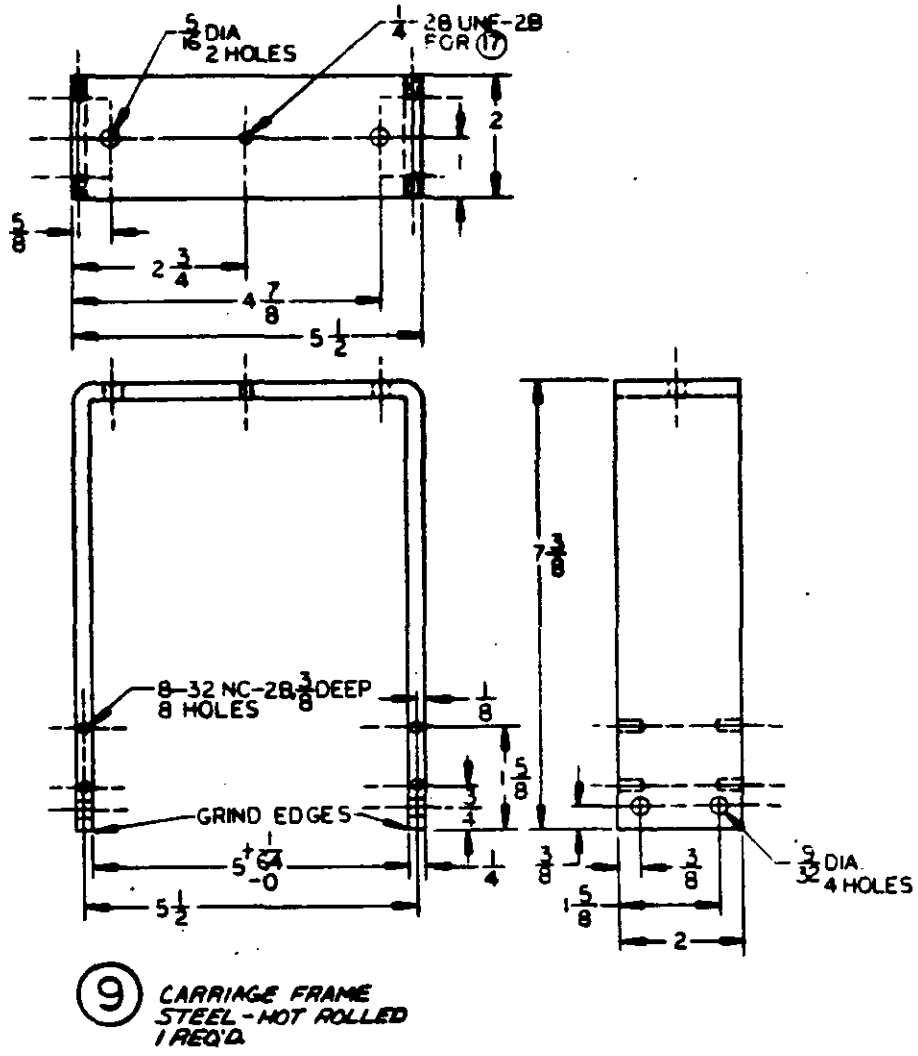


FIGURE 4. Carriage details (see fig. 5).

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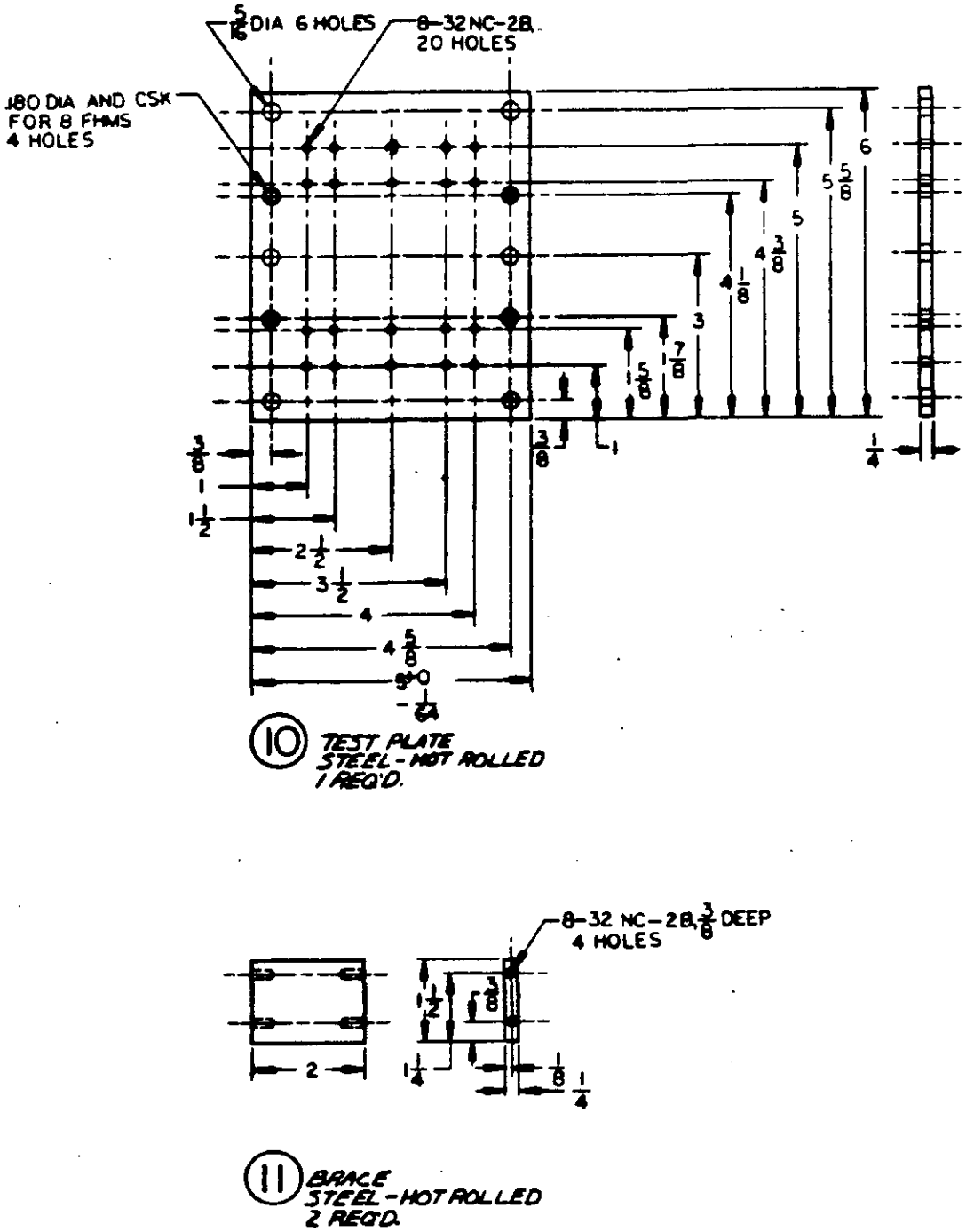


FIGURE 4. Carriage details (see fig. 5). -cont'd

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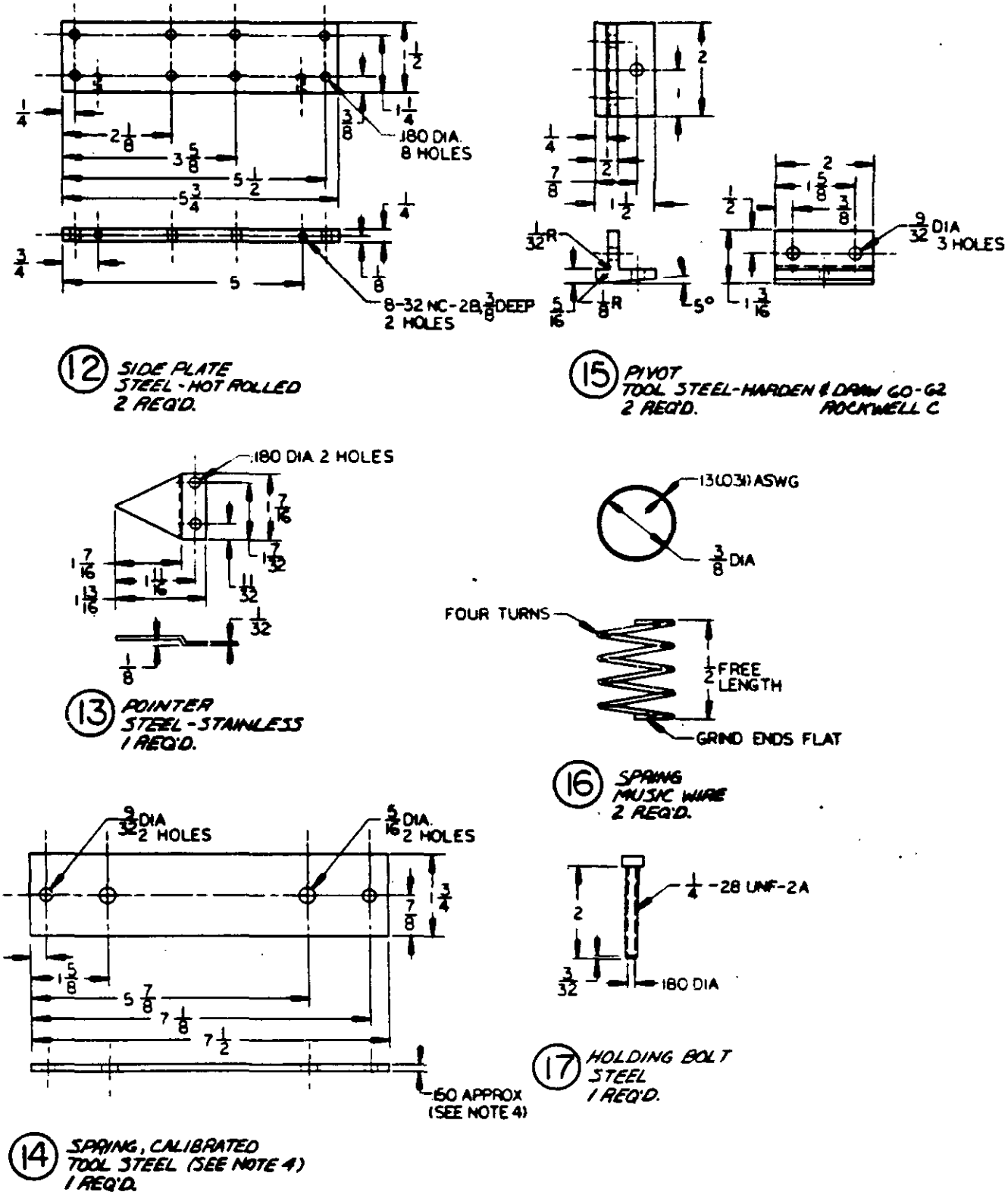


FIGURE 4. Carriage details (see fig. 5). -cont'd

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Item No.	Description	Quantity
1	Left-side channel	1
2	Right-side channel	1
3	Scale	1
4	Guide rod	2
5	Top channel	1
6	Anvil	1
7	Top angle	2
8	Base	1
9	Carriage frame	1
10	Test plate	1
11	Brace	2
12	Side plate	2
13	Pointer	1
14	Spring, calibrated	1
15	Pivot	2
16	Spring, compression	2
17	Squarehead bolt, No. 1/4-28 UNF-2A x 2 long, steel, modified	1
18	Acorn nut, No. 1/4-20 UNC-2B, steel	2
19	Hexagon nut, No. 1/4-20 UNC-2B, steel	14
20	Flat washer, 1/4, steel	4
21	Stop nut, per note 5	2
22	Screw, MS35206-242	2
23	Screw, MS35206-244	16
24	Screw, MS35190-51	4
25	Bolt, MS90725-6	20
26	Bolt, MS90725-7	4
27	Bolt, MS90725-12	2
28	Lockwasher, MS35338-4	2
29	Lockwasher, MS35338-6	26

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±1/64 on fractions, ±.005 on decimals and ±1/2° on angles.
3. Finish: The general assembly and the carriage shall be given a light spray coat of metal primer and then finished with a full spray coat of lusterless gray enamel. The mounting fixtures shall be cadmium or nickel plated. The spring, part 14, anvil, part 6, and pivots, part 15, shall be flash-dip nickeled.
4. Develop thickness experimentally so that spring constant will be 5,000 to 5,500 pounds-per-inch deflection when hardened and drawn.
5. Stop nut shall be catalog No. 290040, as made by Elastic Stop Nut Corporation, Union, N.J., or equal.
6. The materials specified herein have been found satisfactory, but any suitable material may be used.
7. Scale shall be graduated in tenths of an inch. Fill black.

FIGURE 5. Parts list and notes.

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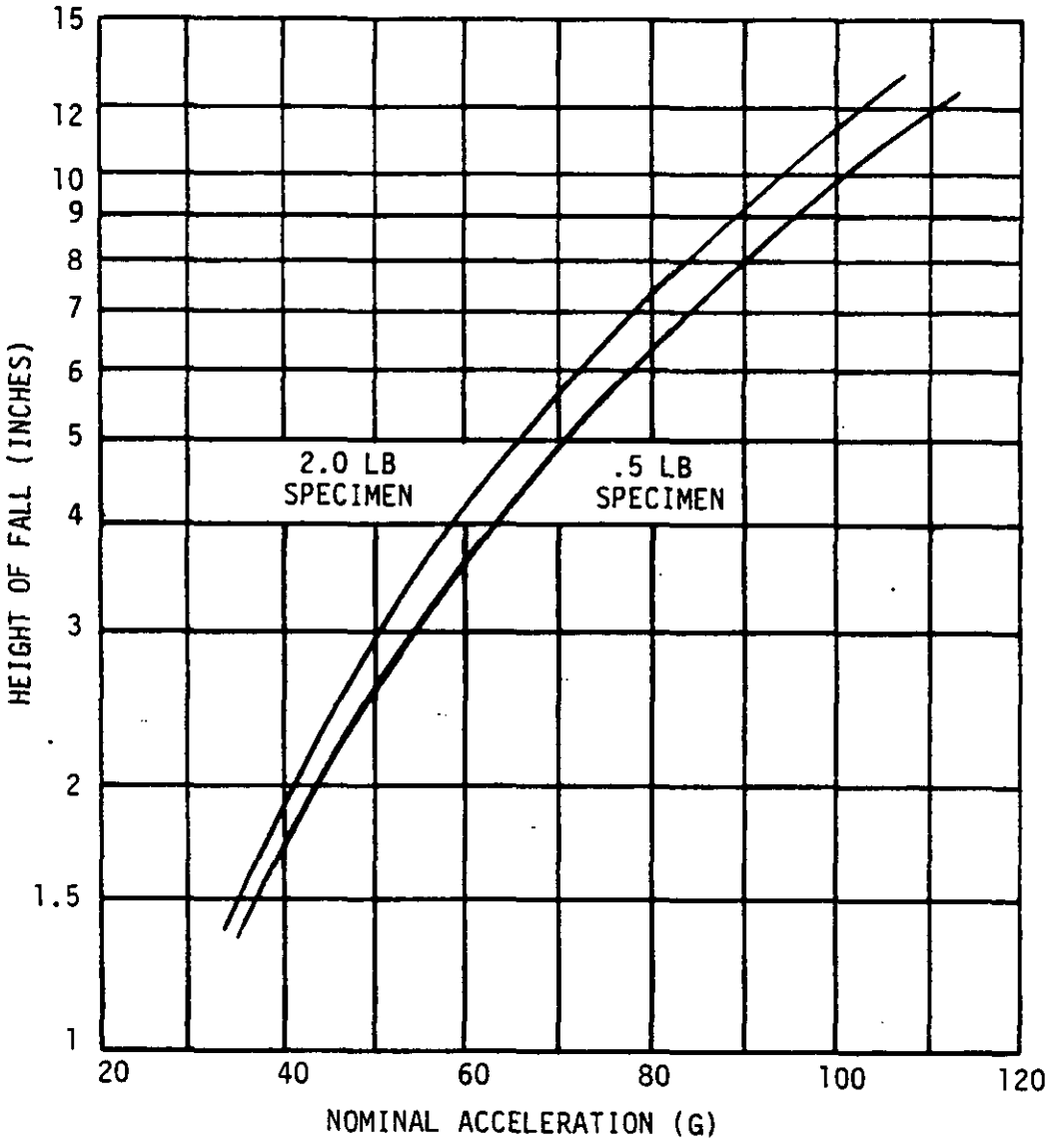


FIGURE 6. Typical calibration curves.

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Custodians:
Army - AT
Air Force - 99

Review activities:
Army - ME
Air Force - 82

User activity:
Navy - MC
Army - CR

Preparing activity:
Army - AT

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	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

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b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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